Instalación HIVE

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1. Creamos el grupo hive

```
hadoop@hadoop-master:~$ sudo groupadd hive
```

2. Creamos usuarios y les añadimos a esos usuarios y al usuario hadoop el grupo hive

```
hadoop@hadoop-master:~$ sudo useradd -d /home/usuarioh1 -m -g hive -G hadoop -s
/bin/bash usuarioh1
hadoop@hadoop-master:~$ sudo useradd -d /home/usuarioh2 -m -g hive -G hadoop -s
/bin/bash usuarioh2
```

```
hadoop@hadoop-master:~$ sudo usermod -G hive hadoop
```

3. Creamos el directorio hive

```
hadoopuser@hadoop-master:~$ sudo mkdir /usr/local/hive
```

4. Cambiamos el grupo del directorio hive a hive

```
hadoopuser@hadoop-master:~$ sudo chown -R hadoop:hive /usr/local/hive/
```

5. Descargamos y descomprimimos hive versión 3.1.3

```
hadoop@hadoop-master:~/Descargas$ tar xvf apache-hive-3.1.3-bin.tar.gz -C /opt/h
ive/
```

6. Movemos todo el contenido a /home/hadoopuser

```
hadoop@hadoop-master:~$ sudo mv /opt/hive/* /home/hadoopuser/
```

7. Sacamos todo el contenido de la carpeta

```
hadoopuser@hadoop-master:~$ sudo mv /usr/local/hive/apache-hive-3.1.3-bin/* /usr
/local/hive/
```

8. Cambiamos el grupo de todos los archivos de /hive a hive y comprobamos que se ha cambiado bien

```
hadoopuser@hadoop-master:/usr/local/hive$ sudo chgrp -R hive /usr/local/hive/*
```

```
hadoopuser@hadoop-master:/usr/local/hive$ ls -l /usr/local/hive/
total 84
drwxr-xr-x 2 root hive  4096 may  7 20:10 apache-hive-3.1.3-bin
drwxr-xr-x 3 root hive  4096 may  7 20:09 bin
drwxr-xr-x 2 root hive  4096 may  7 20:09 binary-package-licenses
drwxr-xr-x 2 root hive  4096 may  7 20:09 conf
drwxr-xr-x 4 root hive  4096 may  7 20:09 examples
drwxr-xr-x 7 root hive  4096 may  7 20:09 hcatalog
drwxr-xr-x 2 root hive  4096 may  7 20:09 jdbc
drwxr-xr-x 4 root hive 20480 may  7 20:09 lib
-rw-r--r-- 1 root hive 20798 mar 28  2022 LICENSE
-rw-r--r-- 1 root hive   230 mar 28  2022 NOTICE
-rw-r--r-- 1 root hive   540 mar 28  2022 RELEASE_NOTES.txt
drwxr-xr-x 4 root hive  4096 may  7 20:09 scripts
```

9. Cambiamos los permisos de /tmp

```
hadoopuser@hadoop-master:/usr/local/hive$ hdfs dfs -chmod 777 /tmp
```

10. Creamos en Hadoop la carpeta warehouse

```
hadoopuser@hadoop-master:/usr/local/hive$ hdfs dfs -mkdir -p /user/hive/warehouse
```

11. Cambiamos el grupo de /user a hive

```
hadoopuser@hadoop-master:/usr/local/hive$ hdfs dfs -chgrp -R hive /user
```

12. Añadimos todos los siguientes exports en .bashrc

```
hadoopuser@hadoop-master:/usr/local/hive$ sudo nano ~/.bashrc
```

```
  GNU nano 7.2                /home/hadoopuser/.bashrc *
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

export HIVE_HOME=/usr/local/hive
export PATH=$PATH:$HIVE_HOME/bin
export PATH=$PATH:$HADOOP_HOME/bin
export HADOOP_CLASSPATH=$JAVA_HOME/lib/tools.jar
export HIVE_CONF_DIR=/usr/local/hive/conf
export HADOOP_HOME=/usr/local/hadoop

^G Ayuda      ^O Guardar    ^W Buscar     ^K Cortar     ^T Ejecutar   ^C Ubicación
^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea
```

13. Quitamos el .template de todos estos archivos.

```
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo cp hive-default.xml.template h
ive-site.xml
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo cp hive-env.sh.template hive-e
nv.sh
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo cp hive-exec-log4j2.properties
.template hive-exec-log4j2.properties
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo cp hive-log4j2.properties.temp
late hive-log4j2.properties
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo cp beeline-log4j2.properties.t
emplate beeline-log4j2.properties
```

14. Cambiamos el grupo de /hive/conf a hive

```
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo chgrp -R hive /usr/local/hive/
conf/
```

15. Añadimos los 2 exports en hive-env.sh

```
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo nano hive-env.sh
```

```
 GNU nano 7.2                         hive-env.sh *
# fi

# The heap size of the jvm stared by hive shell script can be controlled via:
#
# export HADOOP_HEAPSIZE=1024
#
# Larger heap size may be required when running queries over large number of file>
# By default hive shell scripts use a heap size of 256 (MB).  Larger heap size wo>
# appropriate for hive server.


# Set HADOOP_HOME to point to a specific hadoop install directory
# HADOOP_HOME=${bin}/../../hadoop
export HADOOP_HOME=/usr/local/hadoop
# Hive Configuration Directory can be controlled by:
# export HIVE_CONF_DIR=
export HIVE_CONF_DIR=/usr/local/hive/conf
# Folder containing extra libraries required for hive compilation/execution can b>
# export HIVE_AUX_JARS_PATH=
```

16. Añadimos estos 2 properties en hive-site.xml

```
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo nano hive-site.xml
```

```
 GNU nano 7.2                         hive-site.xml *
  See the License for the specific language governing permissions and
  limitations under the License.
--><configuration>
 <!-- WARNING!!! This file is auto generated for documentation purposes ONLY!
 <!-- WARNING!!! Any changes you make to this file will be ignored by Hive.
 <!-- WARNING!!! You must make your changes in hive-site.xml instead.
 <!-- Hive Execution Parameters -->
 <property>
   <name>javax.jdo.option.ConnectionURL</name>
   <value>jbdc:derby:/usr/local/derby/metascore;create=true</value
 </property>
 <property>
   <name>system:java.io.tmpdir</name>
   <value>/tmp/hive/java</value>
 </property>
```

17. Creamos el directorio /derby y le cambiamos los permisos

```
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo mkdir /usr/local/derby
hadoopuser@hadoop-master:/usr/local/hive/conf$ sudo chmod 777 /usr/local/derby/
```

18. Cambiamos el grupo de /hive a hive

```
hadoopuser@hadoop-master:/usr/local$ sudo chgrp -R hive hive/
```

19. Aplicamos los cambios de .bashrc

```
hadoopuser@hadoop-master:/usr/local$ source ~/.bashrc
```

20. Buscamos la línea 3215 de hive-site.xml para eliminar un description que da error

```
hadoopuser@hadoop-master:/usr/local$ nano +3215 /usr/local/hive/conf/hive-site.x
ml
```

Hay que eliminar el contenido de <description></description>

```
  GNU nano 7.2              /usr/local/hive/conf/hive-site.xml *
  </property>
  <property>
    <name>hive.txn.strict.locking.mode</name>
    <value>true</value>
    <description>
      In strict mode non-ACID
      resources use standard R/W lock semantics, e.g. INSERT will acquire exclu>
      In nonstrict mode, for non-ACID resources, INSERT will only acquire share>
      allows two concurrent writes to the same partition but still lets lock ma>
      DROP TABLE etc. when the table is being written to
    </description>
  </property>
  <property>
    <name>hive.txn.xlock.iow</name>
    <value>true</value>
    <description>

    </description>
  </property>
  <property>

^G Ayuda      ^O Guardar    ^W Buscar     ^K Cortar     ^T Ejecutar   ^C Ubicación
^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea
```

21. Iniciamos la BBDD derby

```
hadoopuser@hadoop-master:~$ schematool -dbType derby -initSchema
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hive/lib/log4j-slf4j-impl-2.17.1.ja
r!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hadoop/share/hadoop/common/lib/slf4
j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Metastore connection URL:        jdbc:derby:;databaseName=metastore_db;create=tr
ue
Metastore Connection Driver :    org.apache.derby.jdbc.EmbeddedDriver
Metastore connection User:       APP
Starting metastore schema initialization to 3.1.0
Initialization script hive-schema-3.1.0.derby.sql
```

22. Ejecutamos hive

```
hadoopuser@hadoop-master:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hive/lib/log4j-slf4j-impl-2.17.1.ja
r!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hadoop/share/hadoop/common/lib/slf4
j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = 6c9c29ec-41f2-4573-82ef-b053aa36b058

Logging initialized using configuration in file:/usr/local/hive/conf/hive-log4j2
.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versio
ns. Consider using a different execution engine (i.e. spark, tez) or using Hive
1.X releases.
hive>
```